

# Apurbo Barua

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## EDUCATION

### University of Arizona

B.S. in Computer Science, Minor in Business Administration | GPA: 3.51/4.00

- **Honors:** Dean's List, Academic Distinction, Wildcat Excellence Scholarship
- **Leadership:** Chairperson of Academic Excellence, Delta Chi Fraternity

Tucson, AZ

Aug 2022 – May 2026

## TECHNICAL SKILLS

**AI/ML:** Generative AI, LLMs, RAG, LangChain, AWS Bedrock, Hugging Face, FAISS, Chroma, XGBoost, SHAP, scikit-learn, TensorFlow, Pandas, NumPy

**Languages:** Python, Java, TypeScript, JavaScript, SQL

**Backend & APIs:** FastAPI, Node.js, Express.js, Django, REST APIs, GraphQL, Microservices

**Frontend:** React, React Native, Expo, HTML, CSS, Tailwind CSS

**Databases & Cloud:** PostgreSQL, MongoDB, DynamoDB, AWS (Lambda, API Gateway, Bedrock, EC2, S3, SAM, IAM)

**DevOps & Systems:** Docker, Git, GitHub Actions (CI/CD), Linux/Unix, Agile/Scrum, CloudWatch

## PROFESSIONAL EXPERIENCE

### Software Engineer Intern

Jan 2026 – May 2026

PwC

Tucson, AZ

- Built an agentic AI chatbot using Claude 3 Haiku via AWS Bedrock with a dual-pass engine that generates and executes live SQL, enabling stakeholders to query workforce data in plain English.
- Developed an XGBoost model with SHAP explainability to predict project extension risk, surfacing operational drivers behind delivery delays for proactive staffing decisions.
- Architected a full-stack workforce planning platform (React, TypeScript, Node.js, AWS RDS PostgreSQL) consolidating 5 enterprise systems into a unified dashboard, deployed on AWS EC2.

### Software Engineer Intern

Aug 2025 – Dec 2025

EduTrend (Quanticore LLC)

Remote

- Designed GraphQL schemas, resolvers, and MongoDB service layers for admin management in a TypeScript mobile backend.
- Built secure authentication and role-based access control using Clerk, supporting OAuth-based login flows for mobile clients.
- Diagnosed and resolved GitHub Actions CI/CD pipeline failures by fixing cross-platform ESLint and Linux build issues, restoring automated testing and deployment.

### Undergraduate Teaching Assistant (CS 110)

Jan 2024 – May 2026

University of Arizona, Dept. of Computer Science

Tucson, AZ

- Built Python automation scripts for grading and testing, reducing manual evaluation time by 10+ hours per week.
- Taught core programming, data structures, and algorithms to 150+ students per semester, debugging Python and Java code in labs and office hours.

## PROJECTS

### UA CS Degree Planner [github.com/ApurboBarua17/RAG\\_chatbot](https://github.com/ApurboBarua17/RAG_chatbot)

- Engineered a production RAG chatbot serving UA CS degree planning queries across 134 vector chunks using Groq (Llama 3.3 70B), FastAPI, and Chroma vector database, replacing a manual advising workflow.
- Designed dual-mode REST API endpoints (JSON + multipart/form-data) with prompt-engineered anti-hallucination guardrails; deployed on Vercel and Hugging Face Spaces via CI/CD.

### Tucson Crime Analysis [github.com/ApurboBarua17/tucson-crime-analysis](https://github.com/ApurboBarua17/tucson-crime-analysis)

- Disproved the assumed late-night arrest peak, revealing daytime as highest-risk, by applying chi-square testing and comparing Logistic Regression vs Random Forest classifiers across 170,000+ Tucson records in scikit-learn.

### Delta Chi Alumni Portal [iamkimball.com](https://iamkimball.com)

- Developed a full-stack alumni portal in Django serving 1,000+ active users, featuring secure registration, admin workflows, and role-based access control.
- Deployed to production on PythonAnywhere with a PostgreSQL backend, maintaining reliable uptime across the active user base.

### Website Status Checker [github.com/ApurboBarua17/Website\\_status\\_checker](https://github.com/ApurboBarua17/Website_status_checker)

- Built a serverless monitoring tool using AWS Lambda, API Gateway, and DynamoDB, exposing RESTful API endpoints for multi-region diagnostics with concurrent Python checks.

## CERTIFICATIONS

**IBM Generative AI Engineering** – IBM / Coursera

2025

**AWS Fundamentals** – Amazon Web Services / Coursera

2025